

DUNMAN HIGH SCHOOL
2023 H2 PHYSICS (YEAR 6)

4	(a)	An ideal gas obeys the equation of state $PV = nRT$ at all temperatures T , volume V , pressure P and number of moles n .				B1
	(b)		work done on gas / J	heat supplied o gas / J	increase in internal energy of gas / J	
		A to B	+360	0	+360 (&)	
		B to C	0	+670	+670 (&)	
		C to D	-810 (&)	0	-810	
		D to A	0 (@)	-220 (@)	-220 (#)	
		&: first, second and third line correct #: -220 correct in right hand column @: other two figures correct in bottom row				B1 B1 B1
	(c)	the gas molecules bounce off the receding piston at lower speeds there is a decrease in kinetic energy of the molecules				B1 B1